

Home Network Topology Overview

The illustration above showcases a standard home network configuration, featuring both wired and wireless elements. This network setup ensures secure internet access, connects various devices, and accommodates different home functionalities.

Essential Components

1. **Internet Connection**
 - This network gains internet access via a service provided by an Internet Service Provider (ISP).
 - The cloud at the top represents the broader internet.
2. **Modem**
 - The modem supplied by the ISP, transforms incoming signals into a format compatible with home network devices.
 - It connects to the router using an Ethernet cable through its WAN port.
 - Usually assigned a dynamic public IP address by the ISP.
3. **Router**
 - Acts as the central hub for managing all network traffic.
 - It offers network Address Translation (NAT) to enable multiple devices to share internet connection, DHCP services to automatically allocate IP addresses to devices, Firewall protection and Wi-Fi access point capabilities for wireless devices.
4. **Network Switch**
 - An 8-port device that enhances wired connectivity.
 - Permits multiple Ethernet devices to connect to the network linked to the router via Ethernet cable.
5. **Wi-Fi Network**
 - Broadcasts on both 2.4 GHz and 5 GHz frequencies.

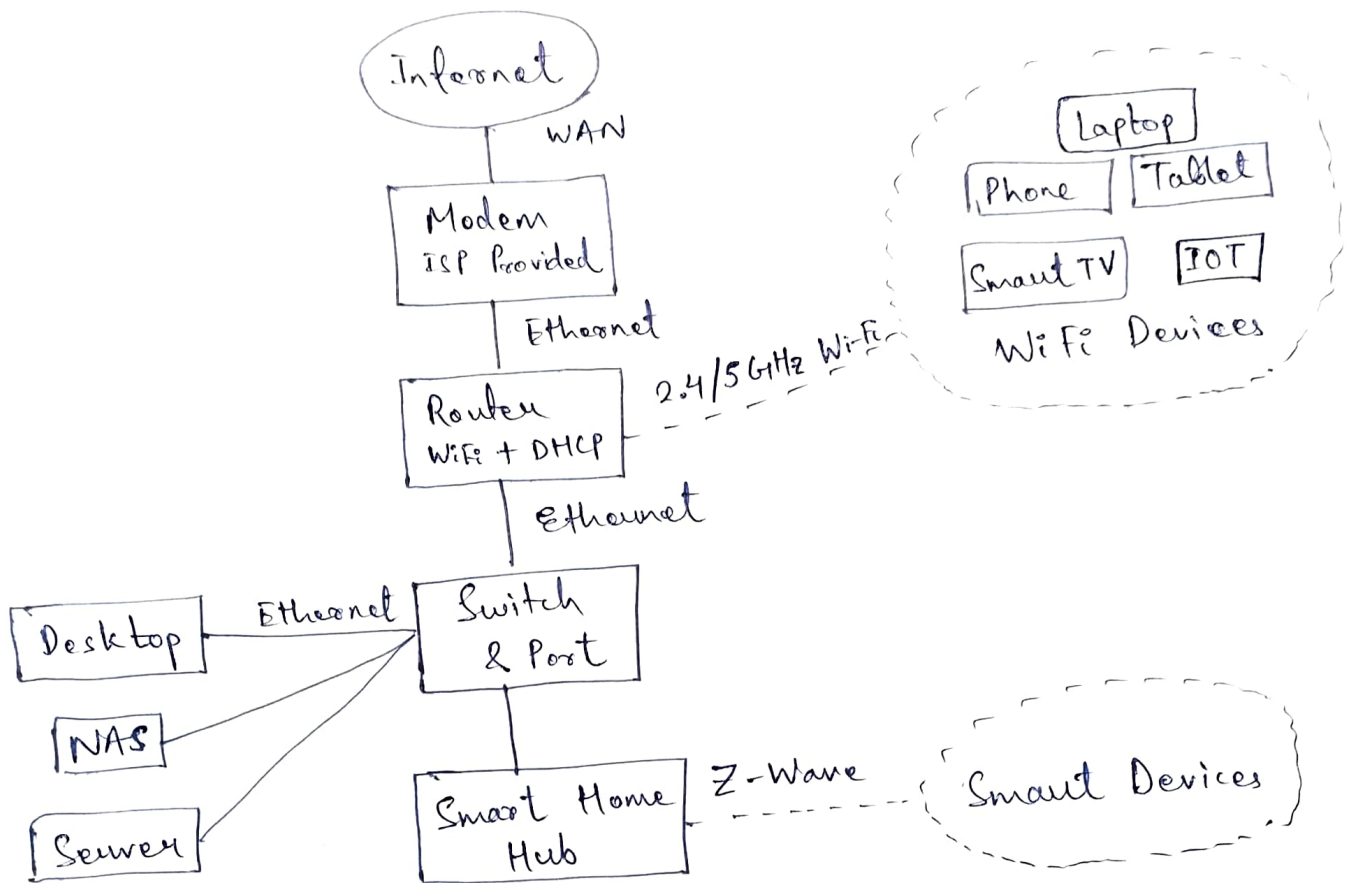
- Uses WPA-3 Personal encryption for security.
- Supports various devices: smartphones, laptop & IoT devices.

6. Wired Devices

- Desktop Computer: Serves as the primary workstation with a reliable, high-speed connection.
- Network Attached Storage (NAS) - facilitates file sharing and backups.
- Home Server - Supports media streaming, development or other services.

7. Smart Home Hub

- functions as the central controller for home automation.
- Connected to the network via Ethernet for a stable connection.
- Communicates with smart devices using protocols.



Network Configuration

WAN IP: DHCP from ISP
 LAN Subnet: 192.168.1.0/24
 Router/Gateway: 192.168.1.1
 DHCP Range: 192.168.1.100-200
 DNS: Router (Primary)
 Wi-Fi: WPA3 - Personal